

Juan David Muñoz-Bolaños

Rechenauerstraße 42, Innsbruck, Austria (Tirol)

+43 676 9115145

juan.munoz@i-med.ac.at

Besi Switzerland AG

Hiring Manager

Hinterbergstrasse 32a

6312 Steinhausen, Switzerland

Subject: Bewerbung als Vision Engineer (Die Attach)

Sehr geehrte Damen und Herren,

Hochleistungsfähige Vision-Lösungen für die Die-Attach-Technologie der nächsten Generation zu entwickeln – genau das ist die Art von Engineering-Herausforderung, für die ich brenne. Als Optik-Ingenieur kurz vor dem Abschluss meiner Promotion bewerbe ich mich als Vision Engineer bei Besi Switzerland, um mit fortschrittlicher Algorithmik und präzisiertem optischen Systemdesign aktiv zur Weiterentwicklung Ihrer Halbleiter-Montagetechnologie beizutragen.

Im Rahmen meiner Promotion an der Medizinischen Universität Innsbruck spezialisierte ich mich auf die Auslegung und Optimierung komplexer optischer Systeme für hochpräzise Metrologie. Ich entwickelte automatisierte Architekturen für Wellenfrontsensorik und Phasenrekonstruktion und verband dabei Hardware-Prototyping mit robuster C++/Python-Steuerungssoftware. Die enge Zusammenarbeit mit Lieferanten wie Thorlabs garantierte dabei höchste Systemzuverlässigkeit – eine Anforderung, die ich auch in der maschinenintegrierten Bildverarbeitung als selbstverständlich betrachte.

Als Junior Machine Vision Engineer bei INTECOL S.A.S. sammelte ich praxisnahe Erfahrung im industriellen Bereich: Ich implementierte Halcon-basierte Bildverarbeitungspipelines für Hochgeschwindigkeits-Qualitätskontrolle auf Glasflaschenlinien. In enger Zusammenarbeit mit einem interdisziplinären Team optimierte ich Beleuchtung und Optik, um kritische Etiketten-Verifizierungsfehler zu beheben und Linienunterbrechungen dauerhaft zu vermeiden.

Ursprünglich aus Kolumbien, habe ich die letzten vier Jahre meiner Promotion in Innsbruck vollständig in den deutschsprachigen Arbeitsraum integriert. Mein Deutsch ist auf B2-Niveau zertifiziert; Englisch beherrsche ich fließend auf professionellem Niveau. Mit dem Abschluss meines Promotionsvertrags im Mai bin ich kurzfristig verfügbar und kann meine bestehende Aufenthaltserlaubnis problemlos in eine Schweizer Arbeitsbewilligung umwandeln.

Ich freue mich darauf, meine Leidenschaft für Vision-Systeme und meine praktische Ingenieursexpertise in Ihr Team in Steinhausen einzubringen – und gemeinsam mit Besi die Zukunft der optischen Montage-Technologie zu gestalten.

Mit freundlichen Grüßen,

Juan David Muñoz-Bolaños

Optical Engineer

Medical University of Innsbruck

Juan David Muñoz-Bolaños

Optical Engineer

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PROFESSIONAL SUMMARY

Optical Engineer with a specialized background in high-precision metrology, adaptive optics, and industrial machine vision. Expert at bridging hardware design with Halcon, C++, and Python control algorithms to create robust, machine-integrated products. Distinguished by strong problem-solving skills and a systematic approach to R and D.

TECHNICAL COMPETENCIES

Optical System Design and Metrology

Extensive expertise in designing and simulating high-precision optical systems, wavefront sensing, and custom illumination for industrial microscopy.

Machine Vision and Algorithmics

Proficient in Halcon, OpenCV, and Python (including JAX for GPU acceleration) for developing high-performance image processing pipelines and real-time automated calibration.

Real-Time Control and FPGA

Experience in hardware-integrated control software and Labview FPGA for high-speed signal processing and coordination of complex optical systems.

R and D Integration

Proven ability to manage the full lifecycle of complex engineering projects with strong problem-solving skills, from technical specification to final qualification.

PROFESSIONAL EXPERIENCE

Medical University of Innsbruck

Innsbruck, Austria

PhD Researcher - Optics and Metrology

Jan 2022 – Present

- **Product-Driven R and D:** Led the end-to-end development of advanced multiphoton microscopy setups, successfully engineering and transitioning complex laboratory experiments into a cohesive, functional product.
- **System Architecture:** Designed and integrated complete optical systems from laser sources to high-sensitivity detectors, actively managing the physical layout and optical path reliability.
- **Software and Automation (C++/Python):** Developed rigorous data acquisition and automated calibration pipelines to coordinate complex measurement protocols across motion axes and electro-optical components.
- **Hardware/Software Integration and Sourcing:** Teamed up with researchers to integrate adaptive hardware and software for high-precision phase retrieval measurements. Managed daily laboratory operations, including designing setups and sourcing optical/mechanical components from vendors like Thorlabs and Edmund Optics to guarantee reliable outcomes for peer-reviewed publications.

Leibniz Institute of Photonic Technology (IPHT)

Jena, Germany

R and D Assistant Researcher (Photonics)

Jun 2020 – Dec 2021

- **Photonics Prototyping:** Designed and constructed a dispersive 1064 nm fiber probe Raman imaging spectrometer; calibrated vibration-based point-scanning prototypes for clinical tissue applications.
- **Data Analysis and AI:** Bridged theoretical R and D with practical laboratory implementation by applying supervised machine learning algorithms to hyperspectral data for streamlined experimental characterization.

INTECOL S.A.S.

Medellín, Colombia

Junior Machine Vision System Engineer

Mar 2018 – Mar 2019

- **Project Management and Architecture:** Managed the physical design of optical inspection setups, encompassing the analysis and selection of sensors, industrial cameras, illumination schemes, and custom optics.
- **Machine Vision Integration:** Implemented robust, industrial-grade image processing pipelines using Halcon for high-speed manufacturing environments. Successfully solved critical line-stoppage issues by engineering custom illumination systems and optimizing light intensity to reliably highlight labels on fast-moving glass bottles.

EDUCATION

Medical University of Innsbruck

Ph.D. in Adaptive Optics & Microscopy

Austria

Jan 2022 – Present

Friedrich Schiller University Jena

M.Sc. in Photonics

Germany

2019 – 2021

University of Cauca

B.Sc. in Physics Engineering

Colombia

2012 – 2018

PUBLICATIONS & AWARDS

- **Awards and Selections:** Selected for ASML and ZEISS Summer Schools; awarded COLFUTURO Grant 2026 for the best Colombian professionals.
- Muñoz-Bolaños, J. D., et al. Design of a dispersive 1064 nm fiber probe Raman imaging spectrometer and its application to human bladder resectates. *Applied Sciences*, 14(11), 4726 (2024).
- Muñoz-Bolaños, J. D. et al. Confocal Raman Microscopy with Adaptive Optics. *ACS Photonics* 12(1), 176–184 (2025).
- Sohmen, M., Muñoz-Bolaños, J. D., et al. Optofluidic adaptive optics in multi-photon microscopy. *Biomedical Optics Express* 14(4), 1562–1578 (2023).

LANGUAGES

English (Fluent / Professional),
German (B2 - Working Proficiency),
Spanish (Native)

REFERENCES

Prof. Monika Ritsch-Marte

Head of Institute for Biomedical Physics

monika.ritsch-marte@i-med.ac.at

Medical University of Innsbruck

Dr. Christoph Krafft

Group leader: Spectroscopy Imaging

christoph.krafft@leibniz-ipht.de

Leibniz Institute of Photonic Technology

AGREEMENT FOR GUESTS

to carry out a practical training
with

Mr. **Juan David Munoz Bolanos** Date of birth: **1994-06-04**
temporarily living at **Emil-Wölk-Str. 7, 07747 Jena** (guest),

Student of the Friedrich-Schiller-University Jena, Abbe School of Photonics, gets in execution of his practical course in the research department Spectroscopy and Imaging, working group Raman and Infrared Histopathology (0104) from 2020-06-02 up to 2020-12-31 admission to the necessary rooms within Leibniz-IPHT.

The use of further devices, media and other infrastructure takes place in agreement with the Heads of the respective departments.

Mr. Juan David Munoz Bolanos has to observe the security regulations for working and other regulations valid in Institute as well as instructions of the laboratory and clean room responsible persons of the Leibniz-IPHT.

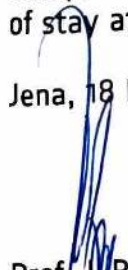
Mr. Juan David Munoz Bolanos has to treat all affairs and procedures strictly confidential. Copies, duplicates and others of business and technical information of each kind are forbidden, unless these are sanctioned by the responsible Department Heads.

Publications, reports and lectures about topics which belong together with the fields of work of Leibniz-IPHT require the previous written agreement of the Executive Board.

The liability of Mr. Juan David Munoz Bolanos is limited to cases of wilful action and gross negligence.

Mr. Juan David Munoz Bolaos is insured against accidents on the Friedrich-Schiller-University Jena, Abbe School of Photonics in the context of his/her practical course during his/her length of stay at Leibniz-IPHT.

Jena, 18 May 2020


Prof. Dr. Popp
Chairman
of Leibniz-IPHT


F. Sonderrmann
Executive Member
of Leibniz-IPHT


Juan David Munoz Bolanos
Guest

INTECOL

OPTIMIZING THE RHYTHM OF THE WORLD

Medellin, February 4, 2019

TO WHOM IT MAY CONCERN

Integración Tecnológica Industrial Intecol S.A.S. with Tax ID (NIT) 811.041.410-4 certifies that Mr. JUAN DAVID MUÑOZ BOLAÑOS, identified with Citizenship ID No. 1.061.772.278, has been employed by this company since July 3, 2018, under an indefinite term contract, serving in the position of Junior Project Engineer;

earning a salary of One million five hundred thousand pesos (\$1,500,000) plus three hundred thousand pesos (\$300,000) mobility allowance.

I will gladly address any additional information at the phone number 316 3322.

We appreciate your attention,

INTECOL

NIT. 811.041.410-4 - Tel.: 3163822

LENY ADRIANA RUIZ PINO

Administrative and Financial Director

Integración Tecnológica Industrial Intecol S.A.S.
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URKUNDE

Die Friedrich-Schiller-Universität Jena verleiht durch die
Physikalisch-Astronomische Fakultät
mit dieser Urkunde

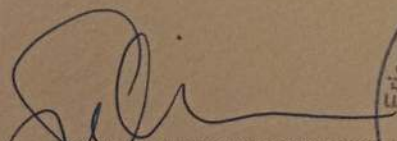
Herrn Juan David Munoz Bolanos
geb. 04.06.1994 in La Cruz (Kolumbien)
den Hochschulgrad

MASTER OF SCIENCE (M.Sc.)

Er hat die Masterprüfung (120 ECTS)
im Studiengang »Photonics«

bestanden.

Jena, 20.12.2021


Prof. Dr. Christian Spielmann
Dekan




Prof. Dr. Silvana Botti
Vorsitzende des
Prüfungsausschusses

OFFICIAL TRANSLATION OF DOCUMENTS WRITTEN IN SPANISH, MADE 20
DECEMBER 2018 BY JUAN CARLOS OJEDA-GARCES, OFFICIAL TRANSLATOR
BY RESOLUTION No. 1171 ISSUED BY THE MINISTRY OF JUSTICE, REPUBLIC
OF COLOMBIA.



UNIVERSIDAD DEL CAUCA
ON BEHALF OF THE
REPUBLIC OF COLOMBIA
AN BY AUTHORIZATION OF THE MINISTRY OF NATIONAL EDUCATION
CONSIDERING THAT

JUAN DAVID MUÑOZ BOLAÑOS
CC 1.061.772.278 OF POPAYAN

HAS COMPLIED WITH ALL THE STATUTORY AND LEGAL REQUIREMENTS,
GRANTS THE TITLE OF

PHYSICAL ENGINEER

WITH ALL THE RIGHTS, PRIVILEGES AND DIGNITIES THAT AUTHORIZE HIM
FOR THE PROFESSIONAL EXERCISE

POPAYAN, 16 MARCH 2018

REGISTERED IN THE DIPLOMA BOOK No. 082 FOLIO 437 DIPLOMA No. 396
RESOLUTION No. R-266-07 MINUTE No. 14-07

(SIGNED)

THE PRINCIPAL OF THE UNIVERSITY
THE DEAN OF THE FACULTY
THE SECRETARY GENERAL OF THE UNIVERSITY

Juan Carlos Ojeda - Garcés
TRADUCTOR INTERPRETE OFICIAL
RESOLUCIÓN No 1171
MINISTERIO DE JUSTICIA 1997

Juan Munoz Bolanos

has participated in the

**ZEISS European
Summer School**

Lithography Optics &
Semiconductor Technologies

26 - 27 June 2025

ZEISS Semiconductor Manufacturing Technology, Oberkochen, Germany

Thomas Marschner

Dr. Thomas Marschner
Program Manager Funding Projects

Thomas Stammer

Dr. Thomas Stammer
CTO & Head of Product Strategy



Zertifikat

Muñoz-Bolaños JUAN DAVID

geboren am 04.06.1994

in La Cruz / Kolumbien

hat das Modul **Schriftliche Prüfung**

ÖSD Zertifikat B2

am Prüfungszentrum

BFI Tirol Bildungs GmbH

in Innsbruck / Österreich

bestanden



Datum: 16.09.2025

für das ÖSD: Prüfungsvorsitzende/r
Maria VALLAZZA



ÖSD ist Mitglied von



ösd

österreich schweiz deutschland

ID-Nummer: ZB2Ö25s34291

Zertifikat

Juan David MUÑOZ BALAÑOS

geboren am 04.06.1994

in La Cruz / Kolumbien

hat das Modul **Mündliche Prüfung**

ÖSD Zertifikat B2

am Prüfungszentrum

BFI Tirol Bildungs GmbH

in Innsbruck / Österreich

bestanden



Datum: 11.08.2025

für das ÖSD: Prüfungsvorsitzende/r
Maria VALLAZZA



ÖSD ist Mitglied von



ösd

österreich schweiz deutschland



ID-Nummer: ZB2Ö25m40434

Passports № / Passport No

MUNOZ BOLANOS

COLOMBIANA

Sexo / Sex
Lugar de nacimiento / Place of birth

Fecha de expedición / Date of issue

Fecha de vencimiento / Date of expiry

G. ANTIOQUIA

Forma dei tubuli / *Peperidera* e *Aggraderia*

Dear Family

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